

MIRAGE: Adaptive Multi-OS Honeypot System (Pro Architecture)

1. Vision & Objective

MIRAGE is designed as an adaptive deception system that dynamically simulates multiple operating environments to mislead attackers and capture high-quality behavioral intelligence. The focus is not attribution, but deep interaction analysis.

2. System Architecture Overview

The system is divided into three primary layers:

- 1 Honeypot Layer: Handles attacker interaction (Cowrie SSH honeypot + simulated services)
- 2 Orchestration Layer: Backend engine controlling behavior and switching
- 3 Intelligence Layer: Behavioral analysis and logging

3. Core Components

3.1 Cowrie SSH Honeypot

Runs as an isolated container. Captures attacker commands, session data, and interaction logs in JSON format without modifying core logic.

3.2 Deception Engine

Simulates OS-level behavior dynamically. Responds to commands such as 'ls' (Linux) or 'dir' (Windows) to maintain realism.

3.3 Behavioral Analysis Engine

Analyzes attacker inputs including command patterns, timing, and tool usage to classify intent.

4. Data Flow Pipeline

- 1 Attacker connects via SSH
- 2 Cowrie logs all commands and interactions
- 3 Backend consumes logs in real-time
- 4 Behavior engine classifies attacker
- 5 System adapts environment dynamically

5. Technology Stack

- 1 Docker for containerization
- 2 FastAPI for backend orchestration
- 3 JSON logs for storage
- 4 tcpdump for traffic monitoring

6. Local Detection & Monitoring Commands

- 1 netstat / ss → detect open ports
- 2 ps aux / top → monitor processes
- 3 lsof -i → check network connections
- 4 tcpdump → analyze live traffic

7. Known Limitations

- 1 Cannot identify real attacker identity (VPN/TOR)
- 2 Cannot inspect encrypted payloads
- 3 Limited visibility outside honeypot
- 4 Advanced attackers may detect deception

8. Future Enhancements

- 1 AI-based attacker profiling
- 2 Cross-honeypot correlation
- 3 Real-time dashboard visualization
- 4 Automated threat intelligence integration

9. Elevator Pitch

MIRAGE is a next-generation adaptive honeypot system that dynamically alters its environment to deceive attackers while capturing behavioral intelligence. Instead of relying on static defenses, it leverages deception and analysis to transform attacks into actionable insights.

10. Conclusion

MIRAGE provides a scalable and intelligent approach to cybersecurity by combining deception, monitoring, and behavioral analytics into a unified system.